

432:

①

①  $12345678_{10} \rightarrow BC614E_{16}$

$1000000_{10} \rightarrow F4240_{16}$

②  $12345678_{16} \rightarrow 305419896_{10}$

$1000000_{16} \rightarrow 16777216_{10}$

③ A - сырое молоко

B - сыр

C - хлеб

X - сырое молоко и сыр и без хлеба

$X = A \cdot B \cdot \bar{C}$

④

| A | B | $A \rightarrow B$ |
|---|---|-------------------|
| 0 | 0 | 1                 |
| 0 | 1 | 1                 |
| 1 | 0 | 0                 |
| 1 | 1 | 1                 |

Каноническая форма:

$$\begin{aligned}
 A \rightarrow B &= \bar{A}\bar{B} + \bar{A}B + AB \\
 &= \bar{A}(\bar{B} + B) + (A\bar{A})B \\
 &= \bar{A} \cdot 1 + 0 \cdot B \\
 &= \bar{A} + 0 = \bar{A} + B
 \end{aligned}$$

| A | B | $A \leftrightarrow B$ |
|---|---|-----------------------|
| 0 | 0 | 1                     |
| 0 | 1 | 0                     |
| 1 | 0 | 0                     |
| 1 | 1 | 1                     |

Каноническая форма:

$A \leftrightarrow B = \bar{A}\bar{B} + AB$

$$⑥ \quad X = (B \rightarrow A)(\overline{A+B})(A \rightarrow C)$$

②

$$\begin{array}{l} B \rightarrow A = \overline{B} + A \\ A \rightarrow C = \overline{A} + C \end{array} \Bigg| \Rightarrow X = (\overline{B} + A)(\overline{A+B})(\overline{A} + C) =$$

De Morgan

$$X = (\overline{B} + A)(\overline{A} \cdot \overline{B})(\overline{A} + C) =$$

$$= \overline{B}(\overline{B} + A)\overline{A}(\overline{A} + C) = \overline{B} \cdot \overline{A} = \overline{A} \cdot \overline{B}$$

нормальная      нормальная

$$X = \overline{A} \overline{B}$$



$$X = A * B + !A * B + !A * !B$$



| A | B | $A * B$ | $!A * B$ | $!A * !B$ | X |
|---|---|---------|----------|-----------|---|
| 0 | 0 | 0       | 0        | 1         | 1 |
| 0 | 1 | 0       | 1        | 0         | 1 |
| 1 | 0 | 1       | 0        | 0         | 1 |
| 1 | 1 | 0       | 0        | 0         | 0 |



$$X = A * B + !A * !B + !B$$



| A | B | $A * B$ | $!A * !B$ | $!B$ | X |
|---|---|---------|-----------|------|---|
| 0 | 0 | 0       | 1         | 1    | 1 |
| 0 | 1 | 0       | 0         | 0    | 0 |
| 1 | 0 | 0       | 0         | 1    | 1 |
| 1 | 1 | 1       | 0         | 0    | 1 |



$$X = !A * !B + !B + A * B$$



| A | B | $\neg A * \neg B$ | $\neg B$ | $A * B$ | X |
|---|---|-------------------|----------|---------|---|
| 0 | 0 | 1                 | 1        | 0       | 1 |
| 0 | 1 | 0                 | 0        | 0       | 0 |
| 1 | 0 | 0                 | 1        | 0       | 1 |
| 1 | 1 | 0                 | 0        | 1       | 1 |



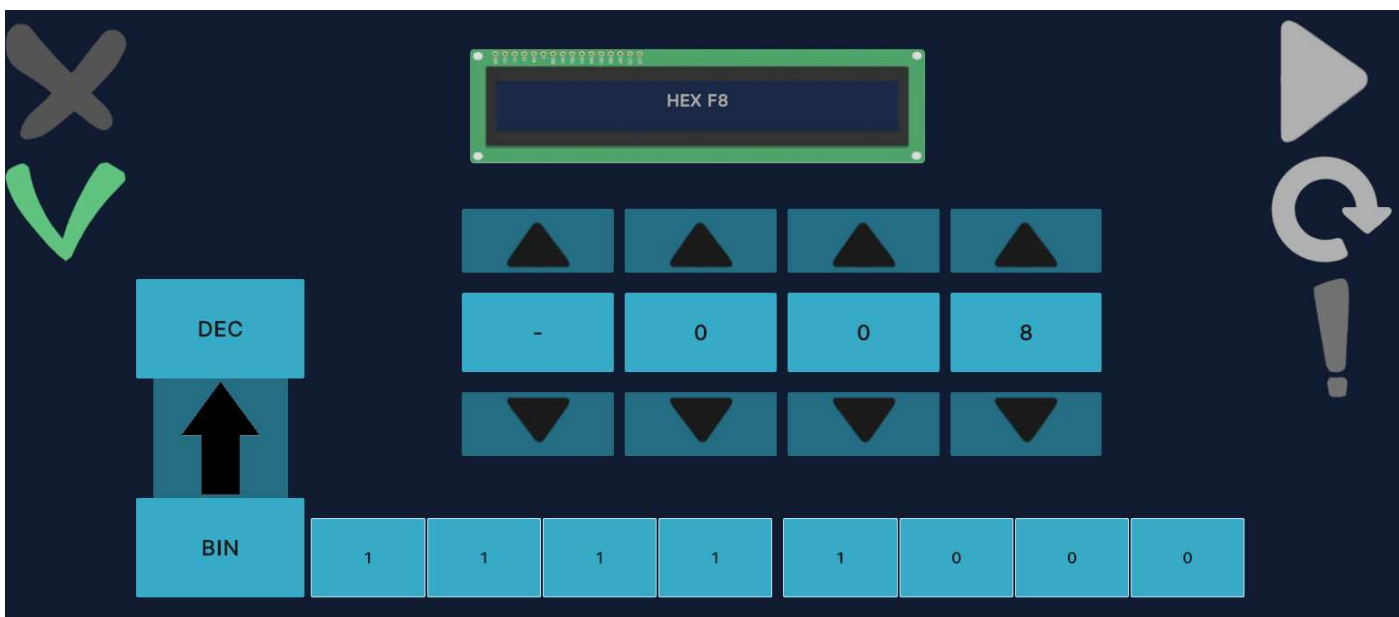
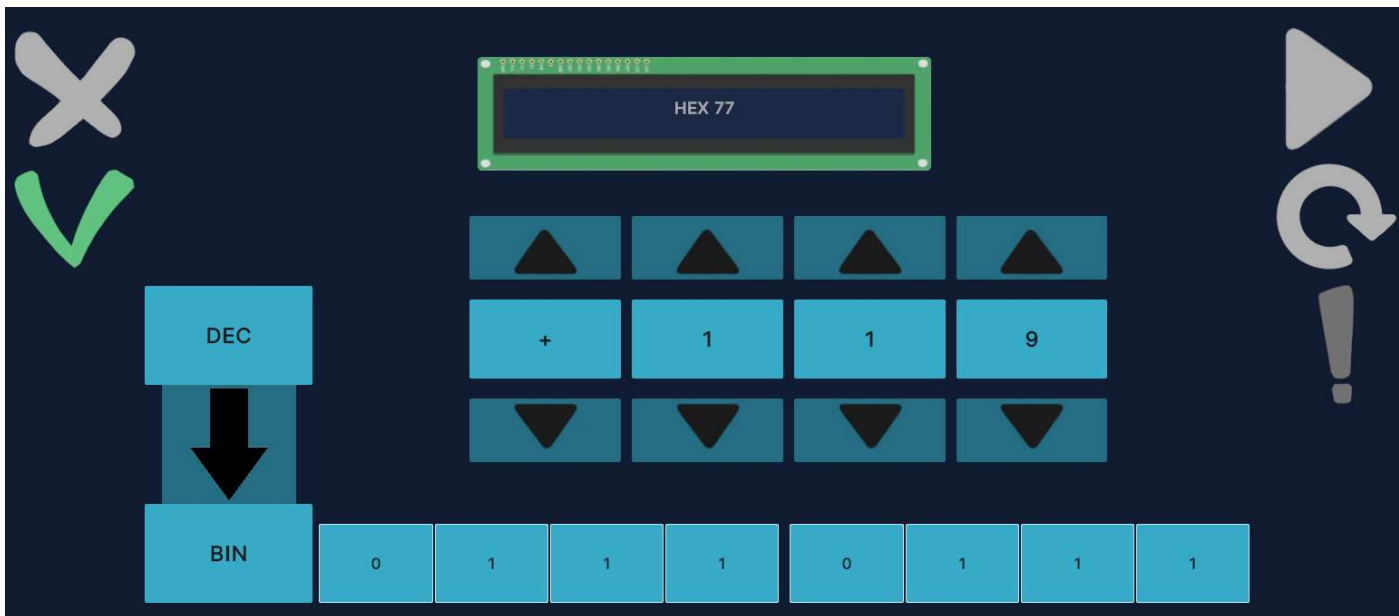
$$X = A * !B + !A + !A * !B$$



| A | B | $A * !B$ | $!A$ | $!A * !B$ | X |
|---|---|----------|------|-----------|---|
| 0 | 0 | 0        | 1    | 1         | 1 |
| 0 | 1 | 0        | 1    | 0         | 1 |
| 1 | 0 | 1        | 0    | 0         | 1 |
| 1 | 1 | 0        | 0    | 0         | 0 |











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HEX 65



DEC



BIN

|   |   |   |   |
|---|---|---|---|
| ▲ | ▲ | ▲ | ▲ |
| + | 1 | 0 | 1 |
| ▼ | ▼ | ▼ | ▼ |

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|